

$$\mathcal{L}[x(t)] = X(s) = \int_0^{\infty} x(t) e^{-st} dt$$

▷ Laplace Transform Properties:

- 1.) $Kx(t) \Leftrightarrow KX(s)$
- 2.) $K_1 x_1(t) + K_2 x_2(t) \Leftrightarrow K_1 X_1(s) + K_2 X_2(s)$
- 3.) $x(at) \Leftrightarrow \frac{1}{a} X\left(\frac{s}{a}\right)$
- 4.) $x(t-T)u(t-T) \Leftrightarrow e^{-Ts} X(s)$
- 5.) $e^{-at} x(t) \Leftrightarrow X(s+a)$
- 6.) $x'(t) \Leftrightarrow sX(s) - x(0^-)$
- 7.) $x''(t) \Leftrightarrow s^2 X(s) - sx(0^-) - x'(0^-)$
- 8.) $\int_0^t x(\tau) d\tau \Leftrightarrow \frac{1}{s} X(s)$
- 9.) $t x(t) \Leftrightarrow -X'(s)$
- 10.) $t^n x(t) \Leftrightarrow (-1)^n X^{(n)}(s)$
- 11.) $x(0) \Leftrightarrow \lim_{s \rightarrow \infty} sX(s)$
- 12.) $\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} sX(s)$
- 13.) $x_1(t) * x_2(t) \Leftrightarrow X_1(s) X_2(s)$

▷ Laplace Transform Pairs:

- 1.) $\delta(t-T) \Leftrightarrow e^{-Ts}$
- 2.) $u(t-T) \Leftrightarrow \frac{1}{s} e^{-Ts}$
- 3.) $e^{-a(t-T)} u(t-T) \Leftrightarrow \frac{1}{s-a} e^{-Ts}$
- 4.) $(t-T)u(t-T) \Leftrightarrow \frac{1}{s^2} e^{-Ts}$
- 5.) $t^2 u(t) \Leftrightarrow \frac{2}{s^3}$
- 6.) $t e^{-at} u(t) \Leftrightarrow \frac{1}{(s+a)^2}$
- 7.) $t^2 e^{-at} u(t) \Leftrightarrow \frac{2}{(s+a)^3}$
- 8.) $t^{(n-1)} e^{-at} u(t) \Leftrightarrow \frac{(n-1)!}{(s+a)^n}$
- 9.) $\sin(\omega_0 t) u(t) \Leftrightarrow \frac{\omega_0}{s^2 + \omega_0^2}$
- 10.) $\sin(\omega_0 t + \theta) u(t) \Leftrightarrow \frac{s \sin(\theta) + \omega_0 \cos(\theta)}{s^2 + \omega_0^2}$
- 11.) $\cos(\omega_0 t) u(t) \Leftrightarrow \frac{s}{s^2 + \omega_0^2}$
- 12.) $\cos(\omega_0 t + \theta) u(t) \Leftrightarrow \frac{s \cos(\theta) - \omega_0 \sin(\theta)}{s^2 + \omega_0^2}$
- 13.) $e^{-at} \sin(\omega_0 t) u(t) \Leftrightarrow \frac{\omega_0}{(s+a)^2 + \omega_0^2}$
- 14.) $e^{-at} \cos(\omega_0 t) u(t) \Leftrightarrow \frac{(s+a)}{(s+a)^2 + \omega_0^2}$
- 15.) $e^{-at} \cos(bt - \theta) u(t) \Leftrightarrow \frac{(s+a) \cos(\theta) + b \sin(\theta)}{(s+a)^2 + b^2}$
- 16.) $\frac{2t^{(n-1)}}{(n-1)!} e^{-at} \cos(bt - \theta) u(t) \Leftrightarrow \frac{e^{j\theta}}{(s+a+jb)^n} + \frac{e^{-j\theta}}{(s+a-jb)^n}$

$$\text{Given } X(s) = \frac{N(s)}{D(s)} \begin{matrix} \leftarrow \text{degree } m \\ \leftarrow \text{degree } n \end{matrix}$$

If $m < n$:

- ↳ Strictly proper
- ↳ Can perform fraction decomp.

If $m \geq n$:

- ↳ Proper ($m=n$) or Improper ($m>n$)
- ↳ Need to divide polynomially
- ↳ Can perform fraction decomp. on remainder term.

▷ Distinct Poles:

$$X(s) = \frac{N(s)}{\prod_{i=1}^n (s-p_i)} = \sum_{i=1}^n \frac{A_i}{(s-p_i)}$$

$$A_i = \left. (s-p_i) X(s) \right|_{s=p_i}$$

↳ If p_i complex:

$$X(s) = \dots + \frac{A_i}{(s-p_i)} + \frac{A_i^*}{(s-p_i^*)} + \dots$$

▷ Repeated Poles:

$$B_k = \left. \left(\frac{1}{(m-k)!} \frac{d^{(m-k)}}{ds^{(m-k)}} ((s-p) X(s)) \right) \right|_{s=p}$$

for $k = 1, 2, \dots, m$

↳ Distinct Complex:

$$\frac{Ae^{j\theta}}{(s+a+jb)} + \frac{Ae^{-j\theta}}{(s+a-jb)} \Leftrightarrow 2Ae^{-at} \cos(bt - \theta) u(t)$$

↳ Repeated Complex:

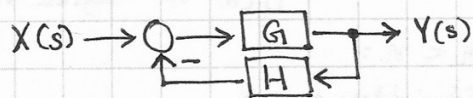
$$\frac{Ae^{j\theta}}{(s+a+jb)^n} + \frac{Ae^{-j\theta}}{(s+a-jb)^n} \Leftrightarrow \frac{2A t^{(n-1)} e^{-at} \cos(bt - \theta)}{(n-1)!}$$

→ System BIBO Stability:

- ↳ Must be properly rational. ($m \leq n$)
- ↳ All poles must have $(-)$ real part
- ↳ System Invertibility:
 - ↳ Must be properly rational ($m=n$)
 - ↳ All poles and zeros have $(-)$ real parts.

▷ Transfer Function Forms:

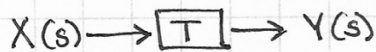
→ Closed Loop Form:



$$\hookrightarrow \frac{Y}{X} = \frac{B}{A} = \frac{G}{1+GH}$$

→ G and H must have overall degree < 0 . Manipulate.

→ Open Loop Form:



$$\hookrightarrow \frac{Y}{X} = \frac{B}{A} \quad \begin{array}{l} \leftarrow \text{degree } m \\ \leftarrow \text{degree } n \end{array}$$

→ B and A must be expanded polynomials.

→ If $m < n$:

→ Strictly proper

→ Reformulate as $\frac{1}{G^* + H}$
(For loop gains)

→ If $m = n$:

→ Biproper

→ Polynomial Division

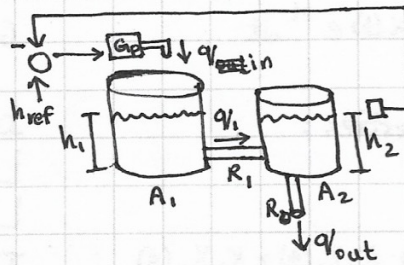
→ Reformulate Remainder

→ If $m > n$:

→ Improper

→ Will not be asked.

▷ Mechanical Describing Diagram:



Given: h_{2s}, h_{1s}

→ A_1, A_2, R_1, R_2, G_p

Asked:

→ $\frac{H_2}{H_{ref}}$

$$\hookrightarrow \frac{Q_{out}}{Q_{in}} = (A_1 A_2 R_1 R_2 s^2 + (A_1 R_1 + A_2 R_2 + A_1 R_2) s + 1)^{-1}$$

▷ Root Locus

→ Start @ L poles

1.) → End @ L zeros

2.) Solutions to char. eq. ($A=0$) are the loci

3.) Locus on real axis will be to left of odd number of zero/pole crossings.

4.) BA: collision of loci on real axis
BI: collision of loci ~~in~~ complex plane

5.) $K \rightarrow \infty$: all poles settle to zeros or a virtual zero @ ∞ .

$$6.) \frac{dL}{ds} = 0 \Rightarrow D_L N_L' - N_L D_L' = 0$$

$$K_{BX} = -\frac{D_L'}{N_L'} \bigg|_{s=BX} \quad \text{or} \quad -\frac{D_L}{N_L} \bigg|_{s=BX}$$

▷ Determining Stability:

→ Routh Array (RA):

→ Recursively find values in a matrix to determine # of unstable poles.

→ Number of sign changes in the leftmost column

→ If there is a 0 in leftmost

→ Use $\epsilon \rightarrow 0^+$ in its place

→ If 2nd row is all 0s:

→ Take derivative and replace the 2nd row

→ "auxillary equation"